

ENGR 102 Summer Bridge

Day 13 Student Quiz

Loop Design, Break, Continue, and Nested Tracing

20 points • approximately 20-25 minutes

Name		UIN		Section	
------	--	-----	--	---------	--

Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace skipped and stopped passes.

5 points

```
score = 1
for value in [3, 0, 4, 8, 2]:
    if value == 0:
        continue
    score = score + value
    if score > 10:
        break
    score = score * 2
print(score)
```

Show score after every reached statement and explain whether 2 is reached.

The following information was obtained from the review of the literature:

Question 2. Repair the skip/stop choice.

3 points

```
for reading in [5, -1, 7, 0, 9]:
    if reading < 0:
        break
    if reading == 0:
        break
print(reading)
```

The rule says negative values are ignored but 0 stops. Give actual output and the minimal repair.

--

Question 3. Program construction

12 points

Write one complete Python program that satisfies every requirement.

- Repeatedly read integer readings. -1 ends normally.
- A reading of 0 is ignored. A reading above 100 stops immediately and is not counted.
- Positive readings from 1 through 100 are counted and added.
- Print count and total. Include one test that contains both an ignored value and an early-stop value.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Test evidence / expected result